

Dylan Spektor

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EDUCATION & HONORS

University of Colorado Boulder

Expected: December 2027

B.S. Applied Mathematics and B.A. Computer Science

- **GPA:** 4.0
- **Organizations:** CU Quants, Engineering Honors Program, Tau Beta Pi, Quantum Scholars.
- **Relevant Coursework:** Mathematical Finance, Probability and Statistics, Algorithms, Numerical Methods and Scientific Computing, Applied Real Analysis, Neural Networks and Deep Learning, Advanced Data Science

Colorado Early Colleges Douglas County North

May 2025

A.S. Computer Science (Dual Enrollment at Arapahoe Community College) | 1580 SAT, National Merit Finalist

RESEARCH & QUANTITATIVE EXPERIENCE

CU Quants Fund

Oct 2025 – Present

Quantitative Developer

- Researching and developing volatility forecasting methods for cryptocurrency market making, with applications to perpetual-futures hedging.
- Design and implement data pipelines over historical orderbook data to support research into stablecoin spread modeling and market-making opportunities.
- Tested crypto lead-lag signals using cross-correlation and transaction costs; prototyped convex capital allocation across pairs strategies.

Undergraduate Researcher

Aug 2026 – Apr 2027

Advised by Prof. Evan Chang, CU Boulder

- Selected for a yearlong research apprenticeship developing algorithmic methods for automated reasoning and formal verification of software systems.
- Building an abstract interpreter in Lean 4; exploring formal verification of AI agents under specification ambiguity.

PROJECTS

[Pairs Trading Strategy Research](#)

April 2026

- Researched and backtested statistical arbitrage across 497 S&P 500 equities (10 years of data), using PCA and weighted PageRank to reduce 124,750 candidate pairs to 300 for Engle-Granger testing; combined GMM regime detection with MinHash pattern matching, then trained a Random Forest achieving 0.97/0.84/0.70 out-of-sample F1 and ~6x baseline cumulative return.

Skills: Python (pandas, NumPy, scikit-learn, statsmodels, datasketch), machine learning, time-series analysis

[SMT-Based Cyclic Arbitrage Detection](#)

July 2026

- Built a Z3 SMT-based cross-currency arbitrage detector supporting fixed fees, nonlinear slippage, and integer lot sizes; validated exact agreement with Bellman-Ford on a frictionless baseline, then uncovered a capital-dependent profitability window (breakeven 0.506; peak profitability at 1.2-1.6x starting capital) missed by Bellman-Ford and confirmed via brute-force enumeration; benchmarked Z3 as ~500x slower on the frictionless case.

Skills: Python, Z3/SMT solvers, graph algorithms, constraint solving, algorithm design, quantitative modeling

[Zeroes of the Riemann Zeta Function and Spectral Statistics of Random Matrices](#)

April 2026

- Co-authored a complex analysis report connecting Riemann zeta zeros to Gaussian/Circular Unitary Ensemble spectral statistics from random matrix theory, exploring the Hilbert–Pólya conjecture. Derived moment-based connections between random matrix and zeta statistics and outlined a contour-integration approach to a key zero-spacing conjecture.

Skills: Mathematical research, statistics, random matrix theory, technical writing & presentation

[External Quantum Efficiency Degradation in Perovskite Cells](#)

December 2025

- An experimental physics research collaboration with the National Laboratory of the Rockies investigating how C60 transport-layer deposition cycles affect perovskite solar-cell EQE degradation; analyzed weekly 350–400 nm EQE measurements across 0, 5, and 10 deposition cycles, using 10,000× bootstrap resampling within measurement uncertainty and quadratic/linear regression to estimate degradation-rate distributions, with ANOVA showing statistically significant differences in both linear and quadratic degradation coefficients across deposition conditions.

Skills: Python (pandas, NumPy, SciPy, scikit-learn), statistical modeling, hypothesis testing, data analysis